Name	Formula	Oxidation state	Physical properties	Preparation
Nitrous oxide	N_2O	**+1**	Colourless gas & neutral	$NH_4NO_3 \rightarrow N_2O + 2H_2O$
Nitric oxide	**NO**	**+2**	Colourless gas & neutral	$2NaNO_2 + 2FeSO_4 + 3H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 2NaHSO_4 + 2H_2O + 2NO$
Dinitrogen trioxide (or) Nitrogen sesquoxide	$\mathbf{N_2O_3}$	**+3**	Blue solid & acidic	$2NO + N_2O_4 \rightarrow 2N_2O_3$
Nitrogen dioxide	$\mathbf{NO_2}$	**+4**	Brown gas & acidic	$2Pb(NO_3)_2 \rightarrow 4NO_2 + 2PbO + O_2$
Nitrogen tetraoxide	$\mathbf{N_2O_4}$	**+4**	Colourless solid & acidic	$2NO_2 \rightarrow N_2O_4$
Nitrogen pentoxide	$\mathbf{N_2O_5}$	**+5**	Colourless solid & acidic	$2HNO_3 + P_2O_5 \rightarrow N_2O_5 + 2HPO_3$

Preparation of nitrogen oxides

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